

Specifications:

Designing a reservation engine for the Arkansas State Park

Implement a function 'calculateStayPrice' with the following parameters:

- int nights
- int guestAge
- boolean isArkansasResident
- boolean hasVeteranDiscount

Rules:

- **Base Price:** \$50/*night*
- **Nights:** Minimum stay is 1 night; maximum stay is 14 nights
- **Age:**
 - **Child (0-12):** 50% off
 - **Adult (13-64):** Full price
 - **Senior (65+):** 20% off
- **Residency & Veteran Status:**
 - AR Residents receive \$10 off total reservation price
 - Veterans receive 10% off reservation price
 - Conflict: If someone is both a resident and a veteran, \$10 applied first, then 10% discount is applied to remainder

→ Analyze Specifications → Identify Causes and Effects

◆ List each distinct **cause** (input condition):

- i. 'nights'

- ii. 'guestAge'
- iii. 'isArkansasResident'
- iv. 'hasVeteranDiscount'

◆ List each distinct **effect** (output/action):

- i. return [double price]

→ Identify Equivalence Partitions

◆ Age Partition(s):

- i. $[0, 12]: price * 0.5$
- ii. $[13, 64]: price$
- iii. $[65, \infty): price * 0.8$

→ Identify Boundary Values

◆ For each partition, pull out edge values: min, max, just-inside, just-outside

◆ Nights BVA:

- i. 0: Exception
- ii. 1: Valid
- iii. 15: Exception
- iv. 14: Valid

◆ Age BVA:

- i. -1: Exception
- ii. 0: $price * 0.5$
- iii. 13: $price$

- iv. 12: $price * 0.5$
- v. 64: $price$
- vi. 65: $price * 0.8$
- vii. Integer.MAX_VALUE: $price * 0.8$

→ Build Decision Table

Cause	Rule #1	Rule #2	Rule #3	Rule #4	Rule #5	Rule #6
isArkansasResident	F	F	F	F	F	T
hasVeteranStatus	F	F	F	F	F	F
age >= 0	T	T	T	T	T	T
age >= 12	T	F	T	T	T	T
age >= 13	F	F	T	T	T	F
age >= 64	F	F	T	F	T	F
age >= 65	F	F	F	F	T	F
Effects	$p = \frac{p}{2}$	$p = \frac{p}{2}$	p	p	$p = p * 0.8$	$p = \frac{p}{2} - 10$

Note: p is a substitute for 'price'

Cause	Rule #7	Rule #8	Rule #9	Rule #10	Rule #11	Rule #12
isArkansasResident	T	T	F	T	T	F
hasVeteranStatus	T	F	T	T	F	T
age >= 0	T	T	T	T	T	T
age >= 12	T	F	F	F	T	T
age >= 13	F	F	F	F	T	T

age >= 64	F	F	F	F	T	T
age >= 65	F	F	F	F	F	F
Effects	$p = (\frac{p}{2} - 10) * 0.9$	$p = \frac{p}{2} - 10$	$p = \frac{p}{2} * 0.9$	$p = (\frac{p}{2} - 10) * 0.9$	$p = p - 10$	$p = p * 0.9$

Cause	Rule #13	Rule #14	Rule #15	Rule #16	Rule #17	Rule #18
isArkansasResident	T	T	F	T	T	F
hasVeteranStatus	T	F	T	T	F	T
age >= 0	T	T	T	T	T	T
age >= 12	T	T	T	T	T	T
age >= 13	T	T	T	T	T	T
age >= 64	T	F	F	F	T	T
age >= 65	F	F	F	F	T	T
Effects	$p = (p - 10) * 0.9$	$p = p - 10$	$p = p * 0.9$	$p = (p - 10) * 0.9$	$p = p * 0.8 - 10$	$p = p * 0.8 * 0.9$

Cause	Rule #19
isArkansasResident	T
hasVeteranStatus	T
age >= 0	T
age >= 12	T
age >= 13	T
age >= 64	T

age >= 65	T
Effects	$p = (p * 0.8 - 10) * 0.9$

➔ Eliminate Duplicates & Prune Illogical Rules

◆ Rules #1 & #2, #3 & #4, #6 & 8, #11 & #14, #12 & #15, #13 & #16 demonstrably have the same effect and may be consolidated

◆ Rules #7, #9, and #10 are illogical: a child should not have a veteran status

◆ Rules #1 - #5 are covered by EPs: removing due to duplication

Cause	Rule #1	Rule #2	Rule #3	Rule #4	Rule #5
isArkansasResident	T	T	F	T	T
hasVeteranStatus	F	F	T	T	F
age >= 0	T	T	T	T	T
age >= 12	T	T	T	T	T
age >= 13	F	T	T	T	T
age >= 64	F	T	T	T	T
age >= 65	F	F	F	F	T
Effects	$p = \frac{p}{2} - 10$	$p = p - 10$	$p = p * 0.9$	$p = (p - 10) * 0.9$	$p = p * 0.8 - 10$

Note: p is a substitute for 'price'

Cause	Rule #6	Rule #7
isArkansasResident	F	T
hasVeteranStatus	T	T
age >= 0	T	T

age >= 12	T	T
age >= 13	T	T
age >= 64	T	T
age >= 65	T	T
Effects	$p = p * 0.8 * 0.9$	$p = (p * 0.8 - 10) * 0.9$

→ Build TCI Table

TCI	Partitions, Values, Rules
EP01	Age Partition I
EP02	Age Partition II
EP03	Age Partition III
BVA01	Nights BVA I
BVA02	Nights BVA II
BVA03	Nights BVA III
BVA04	Nights BVA IV
BVA05	Age BVA I
BVA06	Age BVA II
BVA07	Age BVA III
BVA08	Age BVA IV
BVA09	Age BVA V
BVA10	Age BVA VI
BVA11	Age BVA VII
DT01	Rule #1
DT02	Rule #2
DT03	Rule #3

DT04	Rule #4
DT05	Rule #5
DT06	Rule #6
DT07	Rule #7

→ Build Test Cases

Test Case	TCI(s) Tested	nights	guestAge	isArkansasResident	hasVeteranDiscount	Expected Output
TC01	EP01	1	5	false	false	\$25
TC02	EP02	1	40	false	false	\$50
TC03	EP03	1	70	false	false	\$40
TC04	BVA01	0	30	false	false	Exception
TC05	BVA02	1	30	false	false	\$50
TC06	BVA03	15	30	false	false	Exception
TC07	BVA04	14	30	false	false	\$700
TC08	BVA05	1	-1	false	false	Exception
TC09	BVA06	1	0	false	false	\$25
TC10	BVA07	1	13	false	false	\$50
TC11	BVA08	1	12	false	false	\$25
TC12	BVA09	1	64	false	false	\$50
TC13	BVA10	1	65	false	false	\$40
TC14	BVA11	1	[MAX]	false	false	\$40
TC15	DT01	1	5	true	false	\$15
TC16	DT02	1	40	true	false	\$40
TC17	DT03	1	40	false	true	\$45
TC18	DT04	1	40	true	true	\$36

TC19	DT05	1	70	true	false	\$30
TC20	DT06	1	70	false	true	\$36
TC21	DT07	1	70	true	true	\$27